Jinwei Liu

Ph.D. Student in Control Science and Engineering

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Profile

- Ph.D. student in Control Science and Engineering at the University of Science and Technology of China through a master's-to-Ph.D. transfer, advised by Prof. Yun-Bo Zhao in the Networked Intelligent Control Lab (NICLab).
- During my master's study, I worked with Prof. Pengfei Li.
- Research interests include human-machine hybrid intelligence, human-robot interaction, shared autonomy, and embodied intelligence.
- Current work focuses on shared control, teleoperation, reinforcement learning, and human-in-the-loop reinforcement learning.
- Interested in both human-to-machine intent communication and machine-to-human information expression.

Experience

Dobot, Embodied Intelligence Algorithm Intern — Shanghai, China
June 2026 – present
1 month

Education

Doctor of Philosophy — **University of Science and Technology of China**, Control Science and Engineering — Hefei, China — June 2026 – present
- Master's-to-Ph.D. transfer; combined master's-doctoral track

Master of Engineering — **University of Science and Technology of China**, Artificial Intelligence — Hefei, China — Sept 2024 – June 2026
- Weighted average: 91.09; GPA: 3.99/4.3

Bachelor of Engineering — **Ocean University of China**, Automation — Qingdao, China — Sept 2020 – June 2024
- Rank: 1/89; Weighted average: 92.42; GPA: 3.82/4.0
- Honors: National Scholarship (twice); Honorary Bachelor's Degree of Ocean University of China

Publications

Delay-Aware Shared Control for Teleoperation Systems: Intent Trajectory Prediction Under Communication Latency — 2026
Journal article on delay-aware shared control for teleoperation systems, using DA-RT to predict operator intent under communication latency.
Jinwei Liu, Pengfei Li, Yaqing Zhou, Tao Wang, Yu Kang, Yun-Bo Zhao
ieeexplore.ieee.org/document/11535330

Legible Shared Autonomy: Implicit Communication of Robot Belief through Motion 2026
Accepted paper on legible shared autonomy, enabling robot assistive motion to communicate inferred user goals and confidence through motion.
Jinwei Liu, Pengfei Li, Shaofeng Chen, Tao Wang

LSTM-MPC: Physically Feasible and Interpretable Trajectory Prediction 2026
Accepted paper on physically feasible and interpretable trajectory prediction via latent MPC parameter estimation.
Jinwei Liu, Pengfei Li, Qianqian Zhang, Yunbo Zhao, Yu Kang

Human-AI Shared Control via Motor Knowledge Neuron Stimulation 2026
Accepted paper on human-AI shared control through distributed motor knowledge neuron assemblies.
Jinwei Liu, Pengfei Li, Sibin Huang, Yun-Bo Zhao, Yu Kang